```
(/ocean/projects/cis260028p/nmalik1/env/llm_sys) [nmalik1@v004 llmsys_hw4]$ python -m kernel_tests.test_softmax_bw
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_attn_softmax_bw, ntest [0], dtype [torch.float32]:
(batch_size, nhead, from_len, to_len): (1, 8, 115, 282)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 1.428 / 8.410, speedup: 5.889
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_attn_softmax_bw, ntest [1], dtype [torch.float32]:
(batch_size, nhead, from_len, to_len): (2, 8, 251, 143)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 2.453 / 16.401, speedup: 6.685
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_attn_softmax_bw, ntest [2], dtype [torch.float32]:
(batch_size, nhead, from_len, to_len): (9, 8, 105, 12)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 0.790 / 4.443, speedup: 5.620
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_attn_softmax_bw, ntest [3], dtype [torch.float32]:
(batch_size, nhead, from_len, to_len): (1, 8, 433, 31)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 0.841 / 4.855, speedup: 5.774
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_attn_softmax_bw, ntest [4], dtype [torch.float32]:
(batch_size, nhead, from_len, to_len): (4, 8, 96, 120)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 1.875 / 11.397, speedup: 6.080
```

```
(/ocean/projects/cis260028p/nmalik1/env/llm_sys) [nmalik1@v004 llmsys_hw4]$ python -m kernel_tests.test_softmax_fw
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_attn_softmax, ntest [0], dtype [torch.float32]:
(batch_size, nhead, from_len, to_len, is_dec_self_attn, is_dec_self_attn_infer): (1, 8, 115, 282, False, False)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 1.322 / 9.918, speedup: 7.502
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_attn_softmax, ntest [1], dtype [torch.float32]:
(batch_size, nhead, from_len, to_len, is_dec_self_attn, is_dec_self_attn_infer): (2, 8, 229, 105, False, False)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 1.467 / 13.423, speedup: 9.152
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_attn_softmax, ntest [2], dtype [torch.float32]:
(batch_size, nhead, from_len, to_len, is_dec_self_attn, is_dec_self_attn_infer): (10, 8, 90, 55, False, False)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 1.511 / 13.941, speedup: 9.225
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_attn_softmax, ntest [3], dtype [torch.float32]:
(batch_size, nhead, from_len, to_len, is_dec_self_attn, is_dec_self_attn_infer): (1, 8, 430, 460, False, False)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 3.692 / 46.096, speedup: 12.486
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_attn_softmax, ntest [4], dtype [torch.float32]:
(batch_size, nhead, from_len, to_len, is_dec_self_attn, is_dec_self_attn_infer): (1, 8, 285, 164, False, False)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 1.459 / 12.951, speedup: 8.876
```

```
(/ocean/projects/cis260028p/nmalik1/env/llm_sys) [nmalik1@v004 llmsys_hw4]$ python -m kernel_tests.test_layernorm_fw
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_layernorm, ntest [0], dtype [torch.float32]:
(batch_token_num, hidden_dim): (115, 32)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 0.709 / 40.691, speedup: 57.387
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_layernorm, ntest [1], dtype [torch.float32]:
(batch_token_num, hidden_dim): (502, 32)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 0.740 / 12.389, speedup: 16.744
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_layernorm, ntest [2], dtype [torch.float32]:
(batch_token_num, hidden_dim): (945, 32)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 0.809 / 12.969, speedup: 16.029
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_layernorm, ntest [3], dtype [torch.float32]:
(batch_token_num, hidden_dim): (433, 32)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 0.725 / 12.375, speedup: 17.063
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_layernorm, ntest [4], dtype [torch.float32]:
(batch_token_num, hidden_dim): (384, 32)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 0.708 / 12.181, speedup: 17.199
```

```
(/ocean/projects/cis260028p/nmalik1/env/llm_sys) [nmalik1@v004 llmsys_hw4]$ python -m kernel_tests.test_layernorm_bw
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_layernorm_bw, ntest [0], dtype [torch.float32]:
(batch_token_num, hidden_dim): (115, 32)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 2.678 / 46.632, speedup: 17.414
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_layernorm_bw, ntest [1], dtype [torch.float32]:
(batch_token_num, hidden_dim): (502, 32)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 3.295 / 19.034, speedup: 5.777
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_layernorm_bw, ntest [2], dtype [torch.float32]:
(batch_token_num, hidden_dim): (945, 32)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 4.052 / 20.647, speedup: 5.095
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_layernorm_bw, ntest [3], dtype [torch.float32]:
(batch_token_num, hidden_dim): (433, 32)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 3.197 / 18.819, speedup: 5.886
>>>>>>>>>>>>>>>>>>>>>>>>test_launch_layernorm_bw, ntest [4], dtype [torch.float32]:
(batch_token_num, hidden_dim): (384, 32)
Run baseline...
Run custom...
Compare the results of custom and baseline...
Test passed. Time of custom/baseline (ms): 3.101 / 18.445, speedup: 5.949
```